

## Effect of calixpyrrole in PEO–LiBF<sub>4</sub> polymer electrolytes

M. Kalita<sup>a</sup>, M. Bukat<sup>a</sup>, M. Ciosek<sup>a</sup>, M. Siekierski<sup>a</sup>, S.H. Chung<sup>b</sup>, T. Rodríguez<sup>c</sup>,  
S.G. Greenbaum<sup>d</sup>, R. Kovarsky<sup>e</sup>, D. Golodnitsky<sup>e</sup>, E. Peled<sup>e</sup>,  
D. Zane<sup>f</sup>, B. Scrosati<sup>f</sup>, W. Wieczorek<sup>a,\*</sup>

<sup>a</sup> Warsaw University of Technology, Faculty of Chemistry, Noakowskiego 3, 00-664 Warsaw, Poland

<sup>b</sup> William Paterson University of New Jersey, Department of Chemistry & Physics, Wayne, NJ 07470, USA

<sup>c</sup> University of Puerto Rico Mayagüez Campus, Department of Physics, Mayagüez, Puerto Rico

<sup>d</sup> Hunter College of City University of New York, Department of Physics & Astronomy, New York, NY 10021, USA

<sup>e</sup> School of Chemistry, Tel Aviv University, Tel Aviv 69978, Israel

<sup>f</sup> Dipartimento di Chimica, Università di Roma “La Sapienza”, P.le A. Moro 4, 00185 Rome, Italy

Received 15 October 2004; received in revised form 2 February 2005; accepted 10 February 2005

### Abstract

It is shown that the addition of calix[6]pyrrole to polyether based electrolytes doped with LiBF<sub>4</sub> results in an considerable increase in the cation transport number  $t_{Li^+}$  as confirmed by dc–ac current techniques as well as by PFG NMR studies. The value of  $t_{Li^+}$  in composite electrolytes beyond a certain minimum value weakly depends on the concentration of added calix[6]pyrrole. The increase in lithium transference number is associated with a decrease in ionic conductivity of composite polymeric electrolytes compared to the pure PEO–LiBF<sub>4</sub> systems.

© 2005 Elsevier Ltd. All rights reserved.

**Keywords:** Lithium transference numbers; Anion receptors; Polyethers; Calyx[6]pyrroles

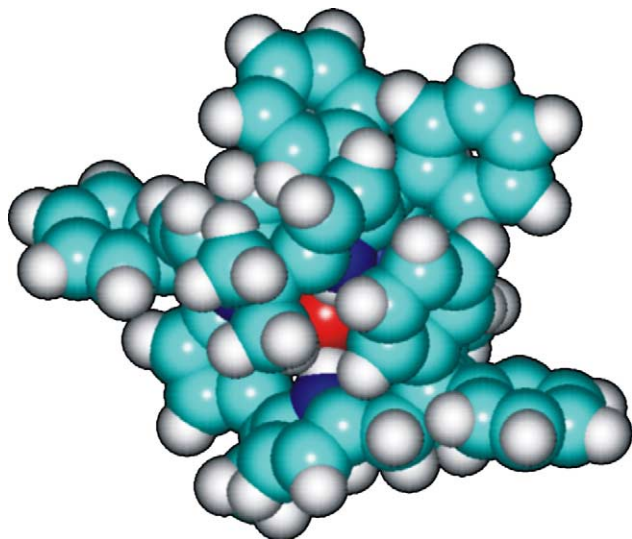
### 1. Introduction

Solid polymer electrolytes based on poly(ethylene oxide) (PEO) have been attracting much attention for application in solvent free lithium rechargeable batteries [1–3]. Both anions and cations are generally mobile in most polymer electrolytes, whereas restricting the mobility of the anions without adversely affecting the lithium cations is desirable for battery applications. Several different strategies have been employed to produce single-ion polymer conductors which, with a transference number of one, avoid the development of concentration gradients that result in low voltage upon discharge. The anion is immobilized usually by covalently connecting it to the polymer matrix. However such lithium single ion polymer conductors have been plagued with low conductivity and mechanical stiffness that leads to poor processing properties [1–3].

Among other methods applied to enhance the cation transport number in polyether based polymeric electrolytes the use of inorganic fillers proved to be one of the most effective as demonstrated by Scrosati's group [4–5]. However, despite an increase in the cation transport numbers the values obtained were still much below unity.

Recently several groups attempt to enhance cation transport number using anionic receptors such as, boron compounds [6], linear or cyclic aza-ether compounds (with electron-withdrawing groups) [7]. Also there are some theoretical predictions which justify the use of anionic receptors as additives enhancing cation transport number in non-aqueous and polymeric electrolytes [8–9]. In our previous papers the use of calix[4]arene derivatives with various types of active groups in a lower rim was shown to be a very effective way to complex iodide and triflate anions and thus giving lithium transport numbers close to unity [10–11]. However, this effect was observed for relatively large fractions of the added calix[4]arene which in hence affect the ionic conductivity of the studied electrolytes

\* Corresponding author. Tel.: +48 22 660 7572; fax: +48 22 628 2741.  
E-mail address: [wladek@soliton.ch.pw.edu.pl](mailto:wladek@soliton.ch.pw.edu.pl) (W. Wieczorek).



Scheme 1. The structure of *meso*-dodekamethylcalix[6]pyrrole-Cl complex [16].

most probably due to the steric hinderance of the bulky receptor.

To further extend our studies on the use of anionic receptors as well as to overcome some of these limitations, we present a different approach involving calix[6]pyrrole as an anion trapping group. Calix[*n*]pyrroles have been identified in various reports as being effective in coordination of different anions [12–16]. As an example of anion complexation the fragment of X-ray structure of *meso*-dodekamethylcalix[6]pyrrole-Cl complex is shown in Scheme 1. It has been also demonstrated that calix[6]pyrrole is a very efficient receptor for  $\text{I}^-$  and  $\text{BF}_4^-$  anions [12].

In the present paper the ionic conductivity as a function of fraction of added calix[6]pyrrole (C6P) was studied by impedance spectroscopy. Lithium transference numbers were investigated by means of ac–dc technique. These studies have been supported by an extensive PFG NMR examination of some of composite systems containing C6P. We investigate using NMR techniques the effectiveness of C6P as an anion trapping agent and hence enhancement of lithium ion transference numbers in the 100:1 PEO– $\text{LiBF}_4$  complex. We report multi-nuclear ( $^1\text{H}$ ,  $^7\text{Li}$ ,  $^{19}\text{F}$ ) NMR self-diffusion coefficient measurements in PEO– $\text{LiBF}_4$  complexes with and without C6P at a temperature above the PEO melt temperature. All diffusing species, cation ( $^7\text{Li}$ ), anion ( $^{19}\text{F}$ ), and polymer or trapping agent ( $^1\text{H}$ ) were measured separately.

## 2. Experimental

### 2.1. Synthesis of calix[6]pyrrole

Anion receptor was synthesized in two-step synthesis proposed by Eichen et al. [12] and improved as follows.

Diphenyldi(pyrrol-2-yl)methane was obtained in condensation of benzophenone with pyrrole.

12 g of pyrrole, 12 g of benzophenone and 12 g of  $\text{BF}_3 \cdot (\text{C}_2\text{H}_5)_2\text{O}$  were dissolved in 120 g of 99.8% ethanol. The reaction mixture was stirred for 5 days at room temperature under argon and then cooled down to  $-5^\circ\text{C}$ . The precipitate that formed was filtered off and washed with 50 ml of cold ethanol and then dried, yielding 10 g of di(phenyl)di(pyrrol-2-yl)methane.

Condensation of di(phenyl)di(pyrrol-2-yl)methane with acetone (catalyst: trifluoroacetic acid) gives 1,1,3,3,5,5-*meso*-hexaphenyl-2,2,4,4,6,6-*meso*-hexamethylcalix[6]pyrrole.

10 g of di(phenyl)di(pyrrol-2-yl)methane and 10 g of trifluoroacetic acid were dissolved in mixture of 1000 g of 99.5% acetone and 1000 g of 99.8% ethanol. The reaction mixture was stirred for 5 days at room temperature under argon. Then, the reaction mixture was cooled down to  $-5^\circ\text{C}$ . The precipitate was filtered off and washed with 50 ml of cold ethanol, and then dried, yielding 2 g of 1,1,3,3,5,5-*meso*-hexaphenyl-2,2,4,4,6,6-*meso*-hexamethylcalix[6]pyrrole. Additional 2 g of 1,1,3,3,5,5-*meso*-hexaphenyl-2,2,4,4,6,6-*meso*-hexamethylcalix[6]pyrrole were obtained by evaporation under vacuum of 1500 ml of the filtrate at  $30^\circ\text{C}$  and filtration of the precipitate. NMR and MS spectra confirmed obtaining the correct product. Before application as additive, anion receptor was dried about 200 h under high vacuum.

### 2.2. Preparation of electrolyte samples

$\text{LiBF}_4$  (Aldrich reagent grade, 99.99%) salt, PEO (molecular weight  $4 \times 10^6$ , Aldrich reagent grade) polymer and C6P, were dried under vacuum at  $100^\circ\text{C}$ ,  $60^\circ\text{C}$  and  $90^\circ\text{C}$ , respectively, for 48 h. Acetonitrile (Aldrich battery grade, 99.93%, water content below 50 ppm)—ACN and dichloromethane (Aldrich biotech grade, 99.9%, water content below 20 ppm) were used as received. All the preparation steps were performed in an argon controlled atmosphere dry-box having a humidity content below 5 ppm.

Lithium salt was quickly (20–30 s) dissolved in ACN at room temperature inside a sealed flask and, then, PEO was added. The solution was carefully stirred in order to dissolve completely the PEO. About 50 ml of ACN are needed for dissolving 1 g of  $4 \times 10^6$  PEO (12–24 h stirring), respectively. C6P was dissolved (30–40 s) in 10–20 ml of dichloromethane at room temperature in a different flask. Then, the two separate solutions were mixed for at least 48 h by magnetic stirring to obtain a homogeneous slurry. No phase separation of the solvents was observed. Successively, the slurry was cast on Teflon® sheets and the solvents were slowly removed (1 day) by room temperature evaporation. Free-standing, homogeneous membrane samples were obtained having a thickness of approximately 100  $\mu\text{m}$  and good mechanical strength. Finally, the membranes were dried under vacuum at  $60^\circ\text{C}$  for at least 48 h.

All electrolytes based on low molecular weight poly(ethylene glycol)dimethyl ether (PEODME) (Aldrich,  $M_w = 500$  g/mol) were prepared by direct dissolution of the lithium salt in the polymer. This was followed by placing an appropriate amount of solution in glass vessels, and the fraction of C6P was added. In order to facilitate dissolving, samples with the highest salt and C6P concentrations were warmed up to 50 °C. All electrolyte samples were equilibrated in a drybox at ambient temperature before being used in any experiments.

### 3. Experimental techniques

#### 3.1. Conductivity measurements

Ionic conductivity at 0–70 °C of liquid and 0–120 °C of solid electrolytes was determined by means of impedance spectroscopy. Liquid electrolytes were placed in a constant-volume cylindrical cell of 1.6 mm-thickness with two symmetrical 7.8 mm-diameter electrodes. Solid electrolytes in the form of discs were pressed between stainless-steel blocking electrodes of 8 mm in diameter. The measurements were carried out on a PC-controlled Solatron–Schlumberger 1255 Impedance Analyzer over the frequency range of 1 Hz–100 kHz.

#### 3.2. Determination of the lithium ion transference number

The lithium ion transference number ( $t^+$ ) of the membrane samples were measured between 50 °C and 90 °C for various content of C6P and salt in the polymer electrolyte system. The steady state technique, which involves a combination of ac and dc-measurements, was applied [17–19]. The impedance response of the Li//solid-electrolyte//Li cell was measured prior to the dc-polarization run, in which a small voltage pulse ( $\Delta V$ ) was applied to the cell until the polarization current reached the steady-state  $I_{ss}$  (after 2 or 3 h at 90 or 50 °C, respectively). Various voltage pulses (3–30 mV) were applied in order to test the experimental error, which did not exceed 10% in determination of transference numbers  $t^+$ . Finally, the impedance response of the cell was measured again. The double ac test is required to determine electrolyte resistance ( $R_e$ ), as well as the solid electrolyte interphase (SEI) resistance before ( $R_0$ ) and after ( $R_{ss}$ ) dc-polarization. That allowed evaluation of the initial current ( $I_0$ ) arising immediately ( $\sim 10 \mu s$  at 50 °C) after appliance of the voltage, since, along with the Ohm's law,  $I_0 = \Delta V / (R_e + R_0)$ .

According to this method, the transference number  $t^+$  of lithium ions, as the carriers of charge flowing through the Li//solid-electrolyte//Li cell, could be calculated with the following equation:

$$t_+ = \frac{I_{ss}(\Delta V - I_0 R_0)}{I_0(\Delta V - I_{ss} R_{ss})} = \frac{R_e}{(\Delta V / I_{ss}) - R_{ss}} \quad (1)$$

Table 1

Lithium transference numbers for PEO–LiBF<sub>4</sub>–calix[6]pyrrole electrolytes

| Type of the electrolyte                 | Molar fraction of calix[6]pyrrole | Lithium transference number |
|-----------------------------------------|-----------------------------------|-----------------------------|
| (PEO) <sub>20</sub> –LiBF <sub>4</sub>  | 0                                 | 0.32                        |
| (PEO) <sub>20</sub> –LiBF <sub>4</sub>  | 0.125                             | 0.78                        |
| (PEO) <sub>20</sub> –LiBF <sub>4</sub>  | 0.25                              | 0.81                        |
| (PEO) <sub>20</sub> –LiBF <sub>4</sub>  | 0.5                               | 0.85                        |
| (PEO) <sub>100</sub> –LiBF <sub>4</sub> | 0.25                              | 0.95                        |
| (PEO) <sub>100</sub> –LiBF <sub>4</sub> | 1                                 | 0.92                        |

Data obtained at 70 °C.

where  $R_0$ : initial SEI resistance of passive layers formed at both lithium electrodes;  $R_{ss}$ : secondary passive layer resistance (as the steady state polarization current is reached).

All the  $t^+$  values obtained from Eq. (1) are collected in Table 1.

#### 3.3. NMR experiments

The NMR measurements were measured on a Chemagnetics CMX-300 spectrometer used in conjunction with a Japan Magnet Technology 7.1 T superconducting magnet. In this field the <sup>1</sup>H ( $I = 1/2$ ), <sup>7</sup>Li ( $I = 3/2$ ), and <sup>19</sup>F ( $I = 1/2$ ) resonances occur at  $\omega_0/2\pi = 301.0$ , 117.0, and 283.2 MHz, respectively.

The pulse gradient spin echo (PGSE) NMR measurements were performed using a 5 mm dual broad band gradient PGSE probe and a current amplifier provided by Nalorac. The multi-nuclear measurements were made using a multiply-tuned Nalorac Z-Spec NMR probe. Experimental details are described elsewhere [20]. The NMR self-diffusion measurements use the Hahn spin-echo pulse sequence [21] with a pair of square-shaped gradient field pulses of magnitude  $g$  and duration  $\delta$ . The first is applied between the two rf pulse and a second identical gradient pulse is applied following the  $\pi$  pulse at a time  $\Delta$  after the first gradient pulse. The echo amplitude is attenuated by an amount dependent on how much the position of the spins has changed by process of self-diffusion in the time interval  $\Delta$ . It can be shown that the attenuation of the spin-echo amplitude is given by

$$A(g) = \exp \left[ -(\gamma g \delta)^2 D \left( \Delta - \frac{\delta}{3} \right) \right] \quad (2)$$

where  $D$  is the self-diffusion coefficient and  $\gamma$  is the magnetogyric ratio of the spin. The diffusion coefficient  $D$  is determined by fitting this equation to the echo amplitudes for a series of gradient strengths  $g$  (0.2–1.2 T/m). Typical reproducibility of the diffusion coefficient measurements is about  $\pm 3\%$ . Thin films of the electrolyte were sealed under Ar ( $< 5$  ppm water) in 5 mm NMR tubes.

### 4. Results

Fig. 1 presents Arrhenius plots of ionic conductivity for (PEO)<sub>100</sub>–LiBF<sub>4</sub> and (PEO)<sub>20</sub>–LiBF<sub>4</sub> electrolytes with and without C6P additives. For both systems the addition of C6P

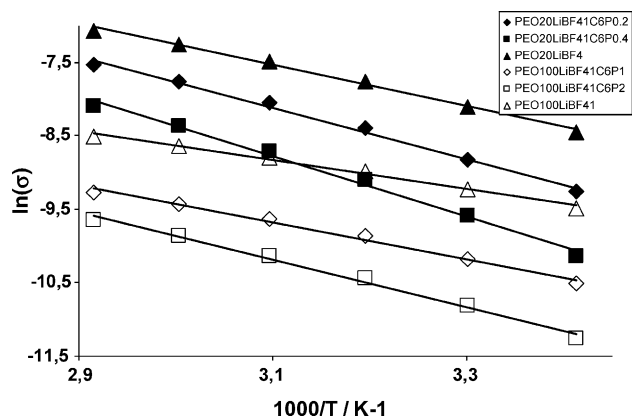


Fig. 1. Arrhenius plots of ionic conductivity for PEO<sub>20</sub>–LiBF<sub>4</sub> electrolytes for systems: without C6P addition (▲), with 0.2 ratio addition (◆) and 0.4 ratio (■) and for (PEO)<sub>100</sub>–LiBF<sub>4</sub> electrolytes for systems: without C6P addition (△), with 1:1 ratio addition (◇) and 1:2 ratio (□).

is followed by a decrease in the conductivity. This decrease is more manifested for composite electrolytes with the high supramolecular additive concentration. Also it can be noticed that, the observed decrease is more pronounced for the (PEO)<sub>100</sub>–LiBF<sub>4</sub> electrolytes.

Table 1 presents values of lithium transference numbers obtained for PEO based electrolytes doped with various amount of LiBF<sub>4</sub> and containing various amounts of C6P. It is particularly well seen for the PEO–LiBF<sub>4</sub> electrolytes that the addition of even small molar fraction (~0.25) of C6P results in the considerable increase in the lithium transference numbers. The further increase in the fraction of C6P results in only small increase in the lithium transference numbers. Also an increase of temperature results in a decrease in the lithium transference number (not shown in the Table 1).

Fig. 2 shows the multinuclear spectra for PEO–LiBF<sub>4</sub> with and without C6P at 90 °C. The <sup>1</sup>H, <sup>7</sup>Li, and <sup>19</sup>F spectra appear as single lines without any structure. The <sup>7</sup>Li and <sup>19</sup>F spectra of the sample containing C6P appear to be slightly broader than without additive. We attribute this to inhomogeneous broadening resulting from the heterogeneous ionic environments due to the presence of C6P, rather than to a dynamic effect. That is, although decreasing the mobility of the anion (the desired result) can cause similar broadening, it turns out that the Li mobility increases, as will be discussed below.

Measurements were possible from 70 °C for <sup>19</sup>F diffusion and 80 °C for the <sup>7</sup>Li and <sup>1</sup>H diffusion. Fig. 3 shows a typical attenuation of the intensity of the <sup>19</sup>F PGSE NMR spin echo spectra at 90 °C of the sample with and without C6P. For a given set of experimental parameters ( $\gamma$ ,  $g$ ,  $\delta$ ,  $\Delta$ ), the slopes of these plots give the diffusion coefficients (see Eq. (2)). The results of the diffusion measurements are summarized in an Arrhenius plot shown in Fig. 4 and in Table 2 which shows the self-diffusion coefficients of the polymer (H<sup>+</sup>), cation (Li<sup>+</sup>), and anion (F<sup>-</sup>) at 90 °C. For both samples, the self-diffusion coefficient follows the order anion > lithium > polymer. To obtain an approximate measure of the cation transference

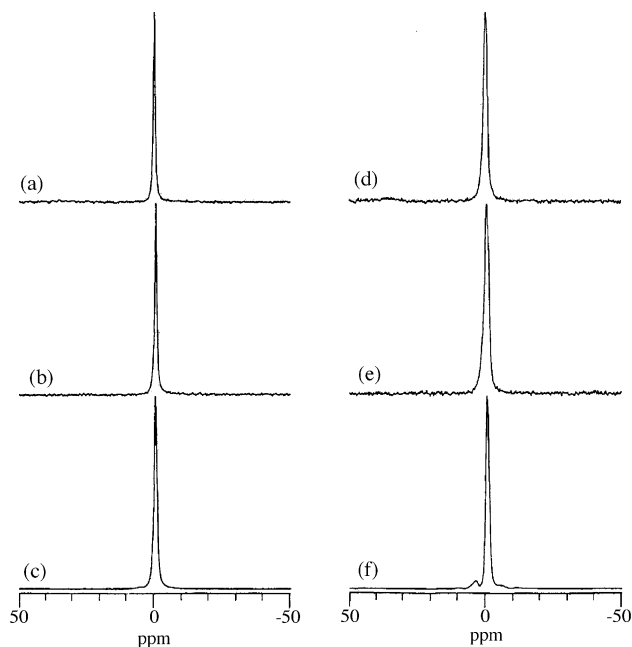


Fig. 2. Lineshapes at 90 °C of 100:1 PEO–LiBF<sub>4</sub>: (a) <sup>19</sup>F—without calixpyrrole, (b) <sup>7</sup>Li—without calixpyrrole, (c) <sup>1</sup>H—without calixpyrrole, (d) <sup>19</sup>F—with calixpyrrole, (e) <sup>7</sup>Li—with calixpyrrole, (f) <sup>1</sup>H—with calixpyrrole.

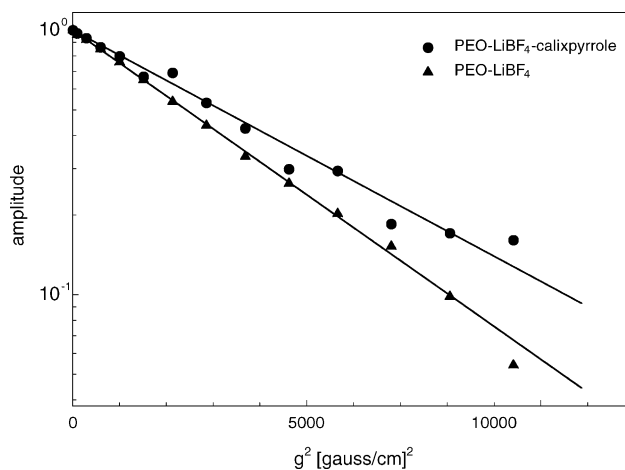


Fig. 3. Signal attenuation plots of <sup>19</sup>F(anion) NMR spin-echo spectra as a function of  $g^2$  at 90 °C in 100:1 PEO–LiBF<sub>4</sub> with and without calixpyrrole.

number  $t^+$  the following relation involving the diffusion coefficients was used:

$$t^+ = \frac{D^+}{D^+ + D^-} \quad (3)$$

Table 2  
Self-diffusion coefficients  $D$  ( $10^{-8}$  cm<sup>2</sup>/s) at 363 K for the mobile species, and the Lithium transference number  $t^+$  as defined by Eq. (2)

|   | $D_{\text{polymer}}$ | $D^-$ | $D^+$ | $t^+$ |
|---|----------------------|-------|-------|-------|
| A | 6.51                 | 27.5  | 24.6  | 0.47  |
| B | 3.37                 | 36.1  | 20.0  | 0.36  |

Samples: A. PEO–LiBF<sub>4</sub>-calixpyrrole, B. PEO–LiBF<sub>4</sub>.

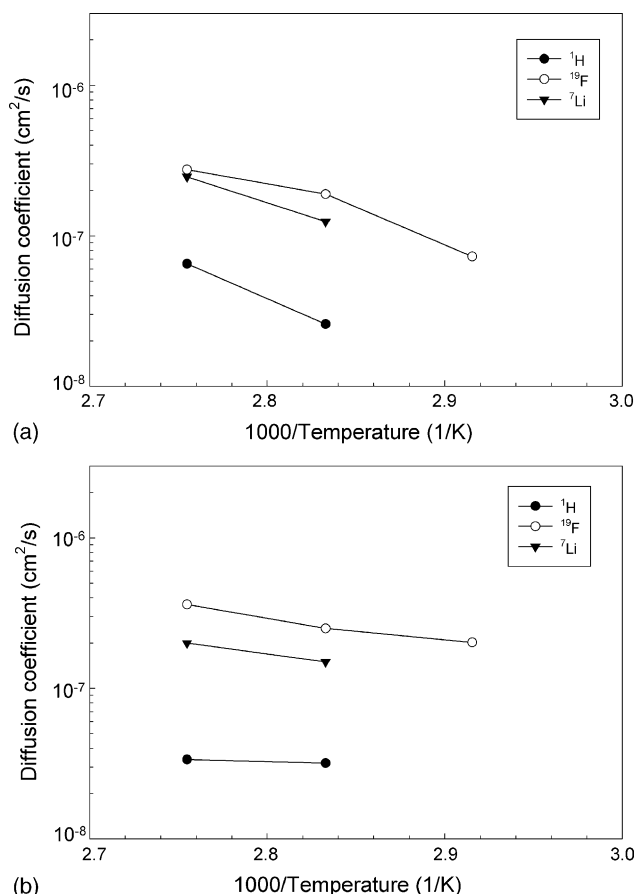


Fig. 4. Arrhenius plot of the self-diffusion coefficients for  $^1\text{H}$ ,  $^7\text{Li}$ , and  $^{19}\text{F}$ . (a) 100:1 PEO-LiBF<sub>4</sub>-calixpyrrole (b) 100:1 PEO-LiBF<sub>4</sub>.

It is believed that  $t^+$  calculated directly from the cation and anion diffusivities is valid for relatively dilute salt concentrations where minimal amount of ion pairing is expected. As it turns even if the absence of ion pairing is a valid assumption, there is some disagreement between two methods of determining cation transport number. This will be addressed later.

The diffusion data show (i) that the anion (i.e.  $\text{BF}_4^-$ ) has a larger diffusion coefficient than the cation in both samples (ii) that the cation (i.e.  $\text{Li}^+$ ) in the sample with C6P has a larger diffusion than in the sample without C6P (iii) that the polymer diffusion (i.e. H) in the sample with C6P has a larger diffusion than in the sample without C6P (iv) that the anion (i.e.  $\text{BF}_4^-$ ) in the sample with C6P has a smaller diffusion than the anion in the sample without the agent.

It is noted that the proton diffusion is significant above the melting point ( $\sim 76^\circ\text{C}$ ), even in the high  $M^W$  ( $\sim 10^6$ ) PEO. The proton results are  $3.4 \times 10^{-8}$  and  $6.5 \times 10^{-8} \text{ cm}^2/\text{s}$  for without and with C6P, respectively. This is consistent with the measured value of pure high  $M^W$  ( $5 \times 10^6$ ) PEO, which was measured to be about  $6 \times 10^{-8} \text{ cm}^2/\text{s}$  at 363 K. Note that the effect of the C6P seems to be a mild plasticizing one (i.e. increasing chain mobility). This result is consistent with a lower glass transition temperature observed for the C6P

containing material. The lithium diffusion is  $2.0 \times 10^{-7}$  and  $2.5 \times 10^{-7} \text{ cm}^2/\text{s}$  without and with C6P, respectively. The lithium diffusion is thus 25% higher from the C6P, but this could possibly be explained by the plasticizing effect mentioned previously.

The anion diffusion data at  $90^\circ\text{C}$  give some insight into the effect of C6P in polymer electrolytes. As noted in Table 2 the fluorine (anion) diffusion is  $3.6 \times 10^{-7}$  and  $2.75 \times 10^{-7} \text{ cm}^2/\text{s}$ , without and with C6P, respectively, about a 30% decrease when the anion trap is added to the polyether-salt system. Our experimental evidence shows that there is definitely significant anion transport, even with the trapping agent. However it can be seen that the cation transference number does indeed improve, from 0.36 to 0.47 when the trapping agent is added to PEO-LiBF<sub>4</sub>.

## 5. Discussion

It is demonstrated that the addition of even relatively small amount (molar fraction =  $\sim 0.25$ ) of C6P to polyether-LiBF<sub>4</sub> electrolytes results in the considerable increase in lithium cation transference numbers as revealed from ac-dc polarization as well as from PFG NMR experiments.

The increase in the lithium transference numbers which is observed upon the addition of C6P is usually associated with a decrease in the conductivity of the samples studied. The simplest explanation of this phenomenon relies on the fact of immobilization of mobile anions and thus a decrease in conductivity. This effect is particularly evident for low concentrations of lithium salt for which high concentration of "free anions" can be expected (see Fig. 1). However, the behavior observed for high salt concentration cannot account on such simple explanation.

Due to the low dielectric constant of polyethers (which can be assumed as a highly viscous or solid solvent) their complexes with lithium salts can be considered as weak electrolytes. Therefore, for high salt concentration range (which is the case of the most systems under study) the presence of large fraction of ion pairs and higher aggregates should be expected. The fraction of ionic aggregates as a function of salt concentration as well as ion pair and triplets formation constants were calculated for LiBF<sub>4</sub> doped electrolytes using Fuoss-Kraus procedure (see Table 3). The ionic triplets formation constant is lower than that of ion pair formation and lower than the receptor-anion complex formation constants which according to the literature data are within the range  $2\text{--}3 \times 10^3 \text{ mol}^{-1}$ . Therefore it can be assumed that it is easier for the receptor to release an anion from triplet than from the ion-pair. On the basis of the literature data [12] the maximum fraction of anion released is equal to molar fraction of reacted C6P equal to 0.25 (based on the DSC data not shown here). Since, according to the literature data that the positive triplets are preferentially formed [22], these implies formation of 0.5 molar fraction of lithium cations which are free to move. The addition of 0.5 molar fraction of released cations



Table 3

Calculated ion association constants and % of salt present as free ions, ion pairs for PEO–DME–LiBF<sub>4</sub> electrolytes

| <i>c</i> (mol/kg)<br>(salt: LiBF <sub>4</sub> ) | % Ions<br>( <i>K</i> <sub>A</sub> : 1.75 × 10 <sup>5</sup> ) | % Pairs<br>( <i>K</i> <sub>T</sub> : 77.69) |
|-------------------------------------------------|--------------------------------------------------------------|---------------------------------------------|
| 9.87E–01                                        | 0.08                                                         | 94.40                                       |
| 7.48E–01                                        | 0.10                                                         | 95.07                                       |
| 5.07E–01                                        | 0.11                                                         | 95.87                                       |
| 2.49E–01                                        | 0.16                                                         | 97.00                                       |
| 9.98E–02                                        | 0.25                                                         | 97.94                                       |
| 5.00E–02                                        | 0.35                                                         | 98.36                                       |
| 2.72E–02                                        | 0.47                                                         | 98.58                                       |
| 1.10E–02                                        | 0.73                                                         | 98.67                                       |
| 5.90E–03                                        | 0.99                                                         | 98.57                                       |
| 8.34E–05                                        | 7.96                                                         | 91.99                                       |
| 4.53E–05                                        | 10.64                                                        | 89.33                                       |
| 3.40E–05                                        | 12.17                                                        | 87.80                                       |

to the lithium transference numbers of pure polymeric electrolytes results roughly in the lithium transference numbers experimentally obtained for composite electrolytes with C6P receptors (see Table 1).

As was already mentioned above also in this salt concentration range, in spite of the increase in cation transport number, the ionic conductivity for composite electrolytes is often much lower than for pure polyether based systems. Besides the reduction in the fraction of mobile anions (which is relatively low in this salt concentration range, see Table 3) this is most probably also due to the steric hindrance of bulky receptor molecules which hindered the ionic transport. In the salt concentration range under discussion the stiffening of the polymer host often results from the formation of inter and intramolecular crosslinks via the positively and negatively charged triplets. The fraction of these aggregated charge carriers is quite high (see Table 3). As was mentioned above the positively charged triplets are preferentially formed. Addition of C6P results in breaking inter and intramolecular crosslinks (due to the release of anion from the positively charged triplet) and therefore increases the flexibility of polyether chains (see PFG NMR data) thus enhancing cation transport. The final decrease or increase in the conductivity results from the above described contradictory effects.

The effect of an increase in lithium transference number observed by PFG NMR is not so large. It is also noteworthy that, although the data in Fig. 4 are limited, the temperature dependence of the diffusion coefficients of the cation and anion are quite similar in the material without C6P, but not so when the anion trap is added. In particular, the cation diffusivity increases more sharply than that of the anion, i.e. the cation transference number increases with increasing temperature. The disagreement between pfg NMR and polarization/steady state methods of determining *t*<sup>+</sup> has several possible origins. The first is that NMR measures all mobile species, including neutral ion pairs and higher aggregates, the former of which do not contribute to the conductivity. A second and more likely scenario is that the ion transport length scales probed by NMR and steady

state method are vastly different, with the former addressing micron-scale motion occurring during the several ms time-frame of the pfg experiment and the latter requiring several hours for the steady state condition to be realized. There is also the question of how the ion trapping interaction may be influenced by the presence of an electric field employed in the steady state method, which is not present during the NMR experiment which probes random motion along the magnetic field gradient axis. These issues, although important, will not be addressed here. Despite the numerical disagreement between methods, the trapping effect is still observed by both.

## 6. Summary

It is shown that the addition of calix[6]pyrrole to polyether based electrolytes doped with LiBF<sub>4</sub> results in an considerable increase in the cation transport number *t*<sub>Li<sup>+</sup></sub> as confirmed by dc–ac current techniques as well as by PFG NMR studies. The value of *t*<sub>Li<sup>+</sup></sub> in composite electrolytes weakly depends on the concentration of added calix[6]pyrrole. The increase in lithium transference number is associated with a decrease in ionic conductivity of composite polymeric electrolytes, however this effect is weaker than in previously studied electrolytes with calix[4]arene.

## Acknowledgements

This research was supported in part by grants from the U.S. Air Force (under the 500?N?1020/0902 research contract). We are grateful to Dr. Stan Bajue for his help with the DSC measurements.

## References

- [1] P.G. Bruce, in: P.G. Bruce (Ed.), *Solid State Electrochemistry*, Cambridge University Press, Cambridge, UK, 1995.
- [2] B. Scrosati, in: B. Scrosati (Ed.), *Applications of Electroactive Polymers*, Chapman and Hall, London, 1993.
- [3] F.M. Gray, in: J.A. Connor (Ed.), *Polymer Electrolytes*, The Royal Society of Chemistry, UK, 1997.
- [4] F. Croce, R. Curini, A. Martinello, L. Persi, F. Ronci, B. Scrosati, R. Caminiti, *J. Phys. Chem.* 103 (1999) 10632.
- [5] F. Croce, G.B. Appetecchi, L. Perci, B. Scrosati, *Nature* 394 (1998) 456.
- [6] (a) H.S. Lee, X.Q. Yang, C.L. Xiang, J. McBreen, L.S. Choi, *J. Electrochem. Soc.* 145 (8) (1998) 2813;  
(b) X. Sun, H.S. Lee, S. Lee, X.Q. Yang, McBreen, *J. Electrochem. Solid State Lett.* 1 (6) (1998) 239.
- [7] F. Zhou, D.R. MacFarlane, M. Forsyth, *Electrochim. Acta* 48 (12) (2003) 1749.
- [8] P. Johansson, *Electrochim. Acta* 48 (14–16) (2003) 2291.
- [9] P. Johansson, P. Jacobsson, this volume.
- [10] A. Blazejczyk, W. Wieczorek, R. Kovarsky, F. Golodnitsky, E. Peled, L.G. Scanlon, G.B. Appetecchi, B. Scrosati, *J. Electrochem. Soc.* 151 (10) (2004) A1762.

- [11] A. Blazejczyk, M. Szczupak, P. Cmocho, W. Wieczorek, R. Kovarsky, F. Golodnitsky, E. Peled, L.G. Scanlon, G.B. Appetecchi, B. Scrosati, *Chem. Mater.* 17 (2005) 1535.
- [12] B. Turner, A. Shterenberg, M. Kapon, K. Suwiska, Y. Eichen, *Chem. Commun.* (2001) 13.
- [13] P.A. Gale, J.L. Sessler, V. Král, *Chem. Commun.* 1–8 (1998), and references cited therein.
- [14] P.A. Gale, P. Anzorbacher Jr., J.L. Sessler, *Coord. Chem. Rev.* 222 (2001) 57, and references cited therein.
- [15] J.L. Sessler, P. Anzorbacher Jr., J.A. Shriver, K. Jursíková, V.M. Lynch, M. Marquez, *J. Am. Chem. Soc.* 122 (2000) 12061.
- [16] G. Cafeo, F.H. Kohnke, G.L. La Torre, A.J.P. White, D.J. Williams, *Chem. Commun.* (2000) 1207.
- [17] P.G. Bruce, C.A. Vincent, J. Electroanal. Chem. Interfacial Electrochem. 225 (1987) 1.
- [18] G. Negasuramanian, E. Peled, A.I. Attia, G. Halpert, *Solid State Ionics* 67 (1993) 51.
- [19] Y. Dai, Y. Wang, S. Bajue, S.G. Greenbaum, D. Golodnitsky, G. Ardel, E. Strauss, E. Peled, *Electrochim. Acta* 43 (1998) 1557.
- [20] S.H. Chung, S. Bajue, S.G. Greenbaum, *J. Chem. Phys.* 112 (2000) 8515.
- [21] E.O. Stejskal, J.E. Tanner, *J. Chem. Phys.* 42 (1965) 288.
- [22] W. Huang, R. Frech, A. Wheeler, *J. Phys. Chem.* 98 (1994) 100.